Veza kontinuiranih i diskretnih signala

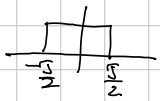
ppt 05

→ signali konačne energije

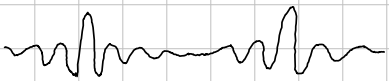
$\int_{-\infty}^{\infty} \sin$ → ne bi trebao imati kon en.

* ali to načelno nisu NUŽNI nego DOVOLJNI UVJETI

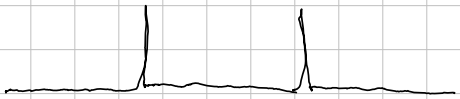
→ odnosno ako je na konačnom segmentu → kon. energija

→ pomnožimo $f(t)$ sa gate fijom →  dobijemo

$\sin(t)$

dva 

* ako se raširi interval gatea → sužava se „iskočni" dio prema 0 a visina (površina) → ∞

 → dobijemo 2 Diraca!

odnosno CTFT:

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\omega)\, e^{j\omega t}\, d\omega \qquad F(j\omega) = \int_{-\infty}^{\infty} f(t)\cdot e^{-j\omega t}\, dt$$

$$F(j\omega) = \delta(\omega) \rightarrow f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(\omega)\cdot e^{j\omega t}\, d\omega = \frac{1}{2\pi}$$

konstanta ide u Diraca!

$$\frac{1}{2\pi} \xrightarrow{CTFT} \delta(\omega)$$

$$e^{jk\omega_0 t} \xleftrightarrow{CTFT} 2\pi\delta(\omega - k\omega_0)$$

$$e^{-jk\omega_0 t} \longleftrightarrow 2\pi\delta(\omega + k\omega_0)$$

$$* \cos(k\omega_0 t) = \frac{e^{jk\omega_0 t} + e^{-jk\omega_0 t}}{2}$$

$$\cos k\omega_0 t \xleftrightarrow{CTFT} \pi\delta(\omega - k\omega_0) + \pi\delta(\omega + k\omega_0)$$

kontinuirani periodični pa ih transformiramo sa CTFS

SPEKTAR SU KOEFICIJENTI RAZVOJA U FOURIEROV RED

ovo je spektar

diskretna domena

$$\cos k\omega_0 t \xrightarrow{CTFS} f(t) = \sum_{n=-\infty}^{\infty} F_k \cdot e^{jn\omega_0 t}$$

n ne je ω (samo je ista oznaka jer je tako formula)

$$F_1 = \frac{1}{2} \qquad F_{-1} = \frac{1}{2}$$

ovo je spektar

ovo je spektar

⟹ vremenski kont. domena

Ako imamo spektar u skupu CTFS, možemo li doći do spektra (isto bilo) u skupu CTFT?

Ako signal imaš rastavljen u FS s koef. k, onda će FT biti CTFT tog FS

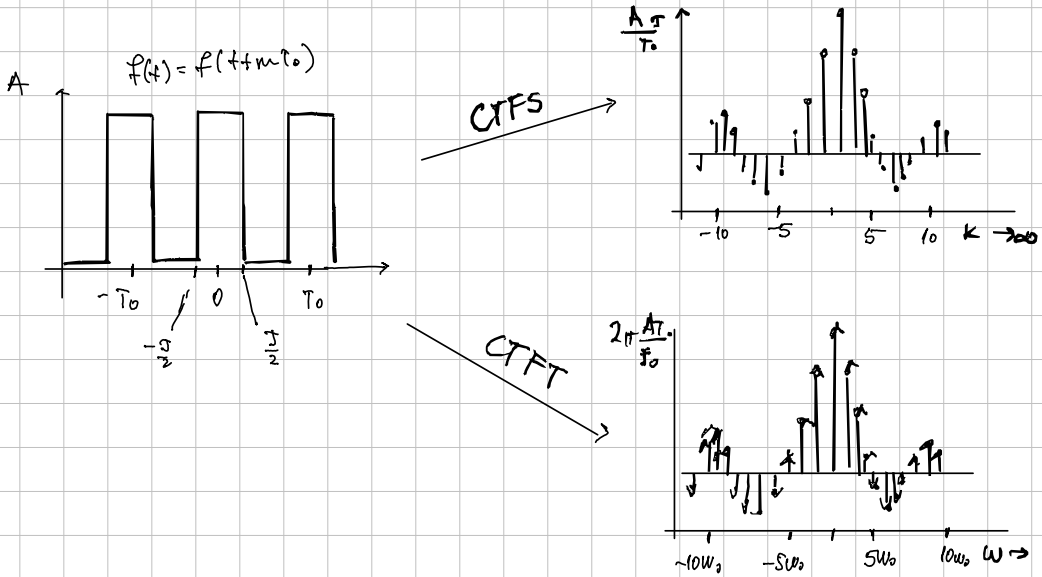
FS. $f(t) = \cos(k\omega_0 t) \xrightarrow{CTFS} f(t) = \underbrace{\sum_{k=-\infty}^{\infty} F_k e^{jk\omega_0 t}}$

i sad kada želim CTFT:

CTFT: $F(j\omega) = CTFT\left\{\sum_{k=-\infty}^{\infty} F_k e^{jk\omega_0 t}\right\} \boxed{= \sum_{k=-\infty}^{\infty} 2\pi F_k \delta(\omega - k\omega_0)}$

„kupimo FT na dobivenu FS"

CTFS i CTFT periodičnog pravokutnog signala



časový funkcyja

$$CTFS\{comb_T(t)\} = F_k = \frac{1}{T} \quad \forall k \in \mathbb{Z}$$

$$CTFT\{comb_T(t)\} = \sum_{k=-\infty}^{\infty} \left(\frac{2\pi}{T}\right) \delta(\omega - k\omega_0) = \sum_k \omega_0 \delta(\omega - k\omega_0)$$

perioda jednej $comb_T(t)$ je isto čo i T_0

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kontinuirani:

$f: \mathbb{R} \longrightarrow \mathbb{R}$

$f(t)$

$f(t+T_0) = f(t)$

$CTFS\{f(t)\} = F_k$

$CTFT\{f(t)\} = \sum_k F_k \, 2\pi \, \delta(\omega - k\omega_0)$

diskretni

$f: \mathbb{Z} \longrightarrow \mathbb{R}$

$f[n+N] = f[n]$

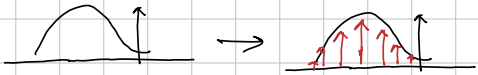
$DTFS\{f[n]\} = F_k$

$DTFT\{f[n]\} = \sum_{k=-\infty}^{\infty} 2\pi F_k \, \delta(\Omega - k\Omega_0)$

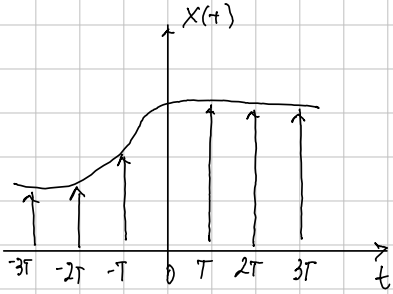
Očitavanje vremenski kontinuiranog signala

uvodimo očitani kontinuirani $\overset{\text{distr.}}{\text{signal}}$ $\rightarrow x_s$

kontinuirani signal × Dirac $\Longrightarrow$



vrijednost signala na kom mjestu × Dirac



$$x[n] = x(t)\Big|_{t=nT}$$

$$\downarrow$$

$$x_s(t) = x(t)\cdot \mathrm{comb}_T(t)$$

$$\Big\downarrow \text{CTFT}$$

20. slaj (20/33)

$$X_s(j\omega) = X(j\omega) * \mathrm{comb}_{\omega_0}(\omega)$$

NOVI SIGNAL

→ periodičan sa ω_s (period impulsiranja Diraca)

Spektar očitanog signala

ako je $\omega_s < 2\omega_{max}$ nije moguća rekonst.

otprije $\mathrm{CTFT}\{\mathrm{comb}_T(t)\} = \frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\delta\left(\omega - k\frac{2\pi}{T}\right)$

$$\mathrm{CTFT}\{x_s(t)\} = X_s(j\omega)$$

$$X_s(j\omega) = \frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\delta\left(\omega - k\frac{2\pi}{T}\right) * \frac{1}{2\pi}X(j\omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty}X\left(j\left(\omega - k\frac{2\pi}{T}\right)\right) = \frac{1}{T}\sum_{k=-\infty}^{\infty}X(j(\omega - k\omega_s))$$

Spektar kontinuiranog očitanog signala je periodičan sa ω_s

$X_s(j\omega - \omega_s)$ (vremenski) → periodičan s ω_s

$X(e^{j\Omega})$ periodičan s 2π